

CSC 2400 Final Project Report

Problem and Goal

The Capacitated Vehicle Routing Problem asks us to assign customers to vehicle routes. Every vehicle leaves the depot, visits customers, and returns to the depot. Each customer is visited once, and the demand on a route cannot exceed the vehicle capacity. Our goal was to compare solution distance, runtime, route count, and basic operation counts.

Algorithms

Nearest Neighbor is the simplest method in the project. It checks every unvisited customer and selects the closest one whose demand still fits. When none fit, it starts another route. Clarke Wright begins with one route for each customer. It calculates the distance saved by connecting every pair. It considers larger savings first and merges routes when the selected customers are endpoints and the combined demand fits.

Harmony Search represents a solution as a customer ordering. The ordering is split whenever the next demand would exceed capacity. A memory stores good orderings. New candidates borrow positions from memory, receive a small swap, and are repaired to contain each customer once. Better candidates replace the worst memory entry. Exact Search tries every customer ordering and every possible route-break pattern on a six

customer dataset. It gives an optimal reference for that small input, but its rapid growth makes it impractical for the main dataset. The code refuses more than eight customers.

Experiment Design

All heuristics use the same ten-customer dataset and capacity of 40. Each is run five times.

Harmony Search uses five fixed seeds because it includes randomness. The other algorithms are deterministic, but repeated runs still show normal timing variation. The exact method uses the tiny dataset. The experiment runner validates each result before saving it. A valid result contains every customer exactly once, does not put the depot inside a route, and stays within vehicle capacity.

Results

The full generated measurements are in the results. On the Ten customer sample, Nearest Neighbor produced a distance of 365.13, Clarke-Wright produced 318.06, and the best Harmony Search run produced 323.82. All three used four routes. The Exact Search result for the separate tiny dataset was 238.91. Since that dataset is different, its distance should not be directly ranked against the ten customer results. Run the experiment runner to reproduce the table. Lower distance is better. The deterministic heuristics return the same distance in every repeated run. Harmony Search returned different distances for its five fixed seeds.

Discussion

Nearest Neighbor is easiest to trace by hand, but each early greedy choice can affect later routes. Clarke Wright considers how much distance a merge saves and can improve on a purely local nearest choice. Harmony Search spends more time

testing possible customer orders. It may find a shorter answer, but it does not guarantee one. The operation counter means something slightly different in each algorithm. For Nearest Neighbor it counts customer checks. Clarke Wright counts savings pairs and route searches. Harmony Search counts new candidates.

Limitations and Next Steps

The main experiment has only ten customers. More CVRPLIB instances of different sizes would provide stronger evidence. Harmony Search settings should also be tested instead of assuming one memory size and iteration count are best. Euclidean coordinates simplify real delivery conditions such as roads and traffic.